

# Role: Logistics Manager (System Owner)

## 1. Module: Organizational Architecture & RBAC

This is the "Administrative Core" where the Manager builds the virtual structure of the business.

- **Multi-Warehouse Provisioning:** \* Ability to create, rename, and geolocate multiple physical hubs.
  - Set operational hours and capacity limits for each warehouse.
- **Role-Based Access Control (RBAC):**
  - **Credential Generation:** Create unique IDs/Passwords for **Warehouse Managers** and **Dispatchers**.
  - **Access Scoping:** Link specific managers to specific warehouses (e.g., *User "John\_WM" only has access to "North-East Hub" data*).
- **System Audit Logs:** A non-editable ledger recording every high-level action (e.g., who changed a driver's pay rate or deleted an order).

## 2. Module: Global Control Tower (Live Dashboard)

The primary interface for real-time system monitoring.

- **Integrated Fleet Map:** \* Utilizes **Google Maps Platform** to display all active vehicles.
  - Live status indicators: Moving (Green), Idle (Yellow), Maintenance (Red).
- **KPI Visualization:** Real-time charts showing:
  - Total Daily Orders vs. Completed Deliveries.
  - System-wide "Delivery Success Rate."
  - Active bottleneck alerts (e.g., "High traffic detected near Hub B").

## 3. Module: Geofencing & Spatial Intelligence

The "Digital Boundary" system for security and automation.

- **Perimeter Management:** \* **Warehouse Fences:** Circular/Polygonal fences around hubs to trigger "Auto-Arrival" and "Auto-Departure" timestamps.
  - **Delivery Territories:** Assigned zones for specific hubs to prevent route overlap and reduce fuel consumption.
- **Security Triggers:** \* **Unauthorized Movement:** Instant alerts if a truck moves outside a geofence without an active delivery manifest.
  - **No-Go Zones:** Temporary barriers created by the Manager to keep drivers out of hazardous or high-traffic areas.
- **Dwell Time Analytics:** Monitoring how long a vehicle stays inside the warehouse geofence to measure loading efficiency.

## 4. Module: Human Capital & Performance Tracking

Managing the people within the system.

- **Driver Scorecards:** \* Aggregated ratings from **Customer Support** feedback.
  - Technical metrics: Average delivery speed, fuel efficiency, and safety incidents.
- **Vendor Portal Oversight:** \* Tracking the performance of 3rd party suppliers.

- Monitoring "Lead Time" from Vendor shipment to Warehouse arrival.

## 5. Module: Financial Ledger & Transactions

The "Economic Engine" of the project.

- **Payment Reconciliation:** \* Monitor **Cash on Delivery (COD)** vs. Digital Pre-payments.
  - Daily reconciliation reports to ensure money collected by drivers matches the system's "Expected Total."
- **Automated Payroll System:** \* Calculating driver earnings based on distance, time, and performance bonuses.
  - Transaction history for every staff member.
- **Expense Auditing:** \* Management of fuel receipts and vehicle maintenance costs uploaded from the field.

## 6. Module: Generative AI Integration (Strategic Layer)

How AI is used specifically for the Manager role to optimize the system.

- **Natural Language Analytics:** \* *Capability:* "Speak to your data." The Manager can ask: *"Who was my most efficient driver in the South Hub last week?"* \* *Action:* AI parses the SQL database and generates a summary and chart.
- **Predictive Optimization:** \* AI analyzes historical data and geofence "Dwell Times" to suggest moving staff between warehouses during peak seasons.
- **Fraud & Anomaly Detection:** \* AI scans financial transactions and GPS logs to flag suspicious behavior (e.g., a fuel claim that doesn't match the GPS-tracked distance).
- **Automated Management Reports:** \* Daily 5-bullet summary generated every morning: "Operational Health: 98%. 1 Delay in Hub A. 3 Drivers due for maintenance."

## 7. Module: Reverse Logistics & Claims Management

This handles the "Backflow" of the system—crucial for costs and customer trust.

- **Return Authorization (RMA):** The Manager oversees the process of a package coming *back* from the customer to the warehouse.
- **Damage Claims Hub:** When a driver or customer reports a damaged item, the Manager can view the "Proof of Damage" photos uploaded via the system.
- **Salvage vs. Restock:** Oversight of the Warehouse Manager's decisions to put an item back into inventory or mark it as a loss.

## 8. Module: Communication & Crisis Management

- **System-Wide Broadcast:** A tool for the Manager to send an "Emergency Alert" to all Dispatchers, Drivers, and Warehouse Managers simultaneously (e.g., "System maintenance at 2 AM" or "Severe weather—all drivers return to base").
- **Internal Ticket System:** A way for the Warehouse Manager or Dispatcher to "Escalate" an issue to the Owner that they cannot solve themselves.

